

HYEYOON JUNG

Federal Reserve Bank of New York
33 Liberty Street
New York, NY 10045, USA

hyeyoon.jung@ny.frb.org
hyeyoonjung.net

EMPLOYMENT AND AFFILIATIONS

Federal Reserve Bank of New York <i>Capital Markets, Research Group (2025- Current)</i> <i>Financial Intermediation, Research Group (2021- 2025)</i>	Jul 2021 – Current Economist
J.P. Morgan (S.E.A) Limited <i>FX & Rates Trading, Currency & Emerging Markets</i>	Jul 2012 – Jul 2015 Associate (FX Trader)

EDUCATION

Leonard N. Stern School of Business, New York University, New York, NY Ph.D. in Finance <i>Dissertation: Essays on International Finance and Financial Stability</i>	2021
M.Phil in Finance	2019
University of Pennsylvania, Philadelphia, PA <i>Jerome Fisher Program in Management and Technology</i>	2012
The Wharton School , Bachelor of Science in Economics, <i>Magna Cum Laude</i>	
The School of Engineering and Applied Science , Bachelor of Applied Science, <i>Magna Cum Laude</i>	

RESEARCH INTERESTS

Asset Pricing, Financial Intermediation, Risk Management, International Finance

WORKING PAPERS

Physical Climate Risk Factors and an Application to Measuring Insurers' Climate Risk Exposure (with Robert Engle, Shan Ge, and Xuran Zeng)

Revise and Resubmit at Review of Financial Studies (RFS)

We construct a novel physical risk factor using a portfolio of REITs, long on those with properties highly exposed to climate risk and short on those with less exposure. Combined with a transition risk factor, we

assess U.S. insurers' climate risk through operations and \$13 trillion in asset holdings. Estimating dynamic climate betas, we find higher sensitivity to physical risk among insurers operating in riskier states and to transition risk among those holding more brown assets. Using these betas, we calculate capital shortfalls under climate stress scenarios, offering insights into insurers' resilience to climate risks.

Economics of Property Insurance (with Jaehoon Kyle Jung)

Best Paper Award, Mid-Atlantic Research Conference in Finance

We study the economics of homeowners' property insurance by examining how contract design balances the trade-off between incentive alignment and risk sharing. Using granular contract-level property insurance data merged with property-level disaster risk for millions of U.S. households, we develop and structurally estimate a model in which insurers optimally determine contract terms given property risk and household risk preferences. The estimates provide, to our knowledge, the first large-scale contract-level structural measures of risk aversion, risk premia, and the cost of moral hazard, allowing us to quantify how disaster risk is allocated between insurers and households. We find that the cost of moral hazard is small, yet the very mechanism used to mitigate it substantially increases households' exposure to disaster risk: contract design leaves policyholders exposed to roughly 29 percent of total expected losses. This residual exposure is most pronounced for low-FICO households and for properties with the greatest tail risk. Counterfactuals indicate that mandating full insurance would lead to substantial market exit, increasing household vulnerability. We further show that insurers' financial constraints are systematically correlated with the riskiness of underwritten properties and with household characteristics.

Credit Card Banking (with Itamar Drechsler, Weiyu Peng, Dominik Supera, and Guanyu Zhou)

Credit card interest rates currently average 22%, an 18% spread over the short rate. This spread far exceeds that on any other loan or bond, yet nearly half of households are credit card borrowers. Why are credit card rates so high? To understand this, and the economics of credit card banking, we use regulatory account-level data to analyze the lifetime cash flows of 550 million monthly accounts, representing 90% of the US credit card market. While charge-off rates are comparatively high, averaging around 6%, they explain only a fraction of cards' spread. Reward payments and non-interest expenses are more than offset by interchange and non-interest income. Operating expenses, particularly marketing, are very large, and are used to generate pricing power. Yet, after deducting them, card lending still earns a 6.8% return on assets (ROA), more than four times the banking sector's ROA. Using the cross section of accounts, we estimate that credit card rates price in a 4.3% default risk premium, similar to high-yield bonds. Accounting for this, card lending earns an alpha of around 1.5% relative to the aggregate bank sector.

Deviations from the Law of One Price across Economies (with Jaehoon Kyle Jung)

In a model with agents facing constraints heterogeneous across economies, we provide a novel explanation for an understudied yet economically significant deviation from the Law of One Price across FX forward markets. Specifically, we document a substantial divergence between the exchange rate for locally traded forward contracts and contracts with the same maturity traded outside the jurisdiction of countries during the global financial crisis, and that the magnitudes varied across currencies. The model predicts that (1) the basis increases with the shadow costs of constraints across time and increases with the country-specific FX position limit across countries; (2) the shadow cost of each constraint non-linearly increases as the intermediary sector's relative performance declines below a threshold; and (3) higher shadow cost of the position limit predicts lower future excess return on local-currency denominated assets, as buying local assets relaxes the FX position limit constraint imposed on the intermediaries. We test the model predictions and find consistent evidence in the countries with tight position limits.

PUBLICATIONS AND FORTHCOMING PAPERS

Real Consequences of Shocks to Intermediaries Supplying Corporate Hedging Instruments

(Job Market Paper)

Review of Financial Studies (RFS), Volume 38, Issue 1, January 2025, Pages 39–113

Financial News & KAFA Top-Journal Paper Award

I show that shocks to financial intermediaries that supply hedging instruments to corporations have real effects. I exploit a quasi-natural experiment in South Korea in 2010, where regulations required banks to hold enough capital for taking positions in foreign exchange derivatives (FXD). Using the variation in exposure to this regulation across banks, I find that the regulation caused a reduction in the supply of FXD, leading to a significant decline in exports for firms that held derivatives contracts with more exposed banks. These results indicate the crucial role of intermediaries in allocating risks through the provision of derivatives and establish a causal relationship between financial hedging and real economic outcomes.

CRISK: Measuring the Climate Risk Exposure of the Financial System (with Robert Engle and Richard Berner)

Journal of Financial Economics (JFE), Volume 171, 2025, 104076

Berkeley Haas Sustainability Research Prize Finalist

We develop a market-based methodology to assess banks' resilience to climate-related risks and study the climate-related risk exposure of large global banks. We introduce a new measure, CRISK, which is the expected capital shortfall of a bank in a climate stress scenario. To estimate CRISK, we construct climate risk factors and dynamically measure banks' stock return sensitivity (that is, climate beta) to the climate risk factor. We validate the climate risk factor empirically and the climate beta estimates by

using granular data on large U.S. banks' loan portfolios. The measure is useful in quantifying banks' climate-related risk exposure through the market risk and the credit risk channels.

U.S. Banks' Exposures to Climate Transition Risks (with Joao Santos and Lee Seltzer)

Journal of Financial and Quantitative Analysis (JFQA)

We propose a new approach to estimate banks' credit exposures to transition risks that combines sectoral effects of climate policies from general equilibrium (GE) models with historical information on loans' default risks. At worst, estimated exposures reach 14% of bank loan portfolio values. Accounting for historic loan payoff structures reduces exposures to 0.5%–2%. Exposures can increase by 3–5 percentage points due to aggregate economic shocks. Analyses surrounding climate transition events suggest our estimates can serve as an upper bound on banks' transition risk exposures. Highlighting our measure's novelty, emissions explain at most 60% of variation in banks' exposures.

Climate Stress Testing (with Viral Acharya, Richard Berner, Robert Engle, Johannes Stroebe, Xuran Zeng, and Yihao Zhao)

Annual Review of Financial Economics 2023 15:1, 291-326

We explore the design of climate stress tests to assess and manage macro-prudential risks from climate change in the financial sector. We review the climate stress scenarios currently employed by regulators, highlighting the need to (i) consider many transition risks as dynamic policy choices; (ii) better understand and incorporate feedback loops between climate change and the economy; and (iii) further explore “compound risk” scenarios in which climate risks co-occur with other risks. We discuss how the process of mapping climate stress scenarios into financial firm outcomes can incorporate existing evidence on the effects of various climate-related risks on credit and market outcomes. We argue that more research is required to (i) identify channels through which plausible scenarios can lead to meaningful short-run impact on credit risks given typical bank loan maturities; (ii) incorporate bank-lending responses to climate risks; (iii) assess the adequacy of climate risk pricing in financial markets; and (iv) better understand and incorporate the process of expectations formation around the realizations of climate risks. Finally, we discuss the relative advantages and disadvantages of using market-based climate stress tests that can be conducted using publicly available data to complement existing stress testing frameworks.

PRESENTATIONS (Including scheduled; Own presentations only)

Seminars: Seoul National University, New York University Stern Quantitative Finance & Econometrics Seminar, Korea University, Federal Reserve Bank of New York, Tinbergen Institute 2026

Conferences: American Finance Association Meeting, ASSA KAEA-KWEN Workshop on International Macroeconomics, Villanova School of Business

Mid-Atlantic Research Conference in Finance, Bocconi-CEPR Finance Workshop on Asset Pricing, Esade Spring Workshop, Baruch-NUS Real Estate Conference, Early Career Women in Finance Conference, System Conference on Financial Institutions, Regulation, and Markets, Lugano Real Estate and Urban Economics Conference

Seminars: Korea University- Korea Advanced Institute of Science and Technology Finance Seminar, Society for Financial Econometrics Summer School, University of New South Wales & University of Sydney Beachside Banking Chats 2025

Conferences: Society for Financial Studies Cavalcade, Financial Intermediation Research Society Conference, European Finance Association Meeting, Global Research Alliance for Sustainable Finance and Investment Conference, Conference in Sustainable Finance at the University of Luxembourg, CEPR-ESSEC-University of Luxembourg Conference, System Conference on Financial Institutions, Regulation, and Markets, WE_ARE_IN Macroeconomics and Finance Conference, University of Toronto Financial Economics Conference, Paris December Meeting

Media: Faculti Academic Media Platform

Seminars: Columbia University, Federal Reserve Bank of New York, E-axes Young Scholar Series on Financial Economics and Sustainability, Financial Stability Oversight Council Meeting 2024

Conferences: American Economic Association Meeting, ASSA Meeting (International Banking, Economics, and Finance Association Session), Midwest Finance Association Meeting, CESifo Conference on Energy and Climate Economics, ZEW Conference on Ageing and Sustainability Finance, Western Economic Association International Meeting (International Banking, Economics, and Finance Association Session), Banca d'Italia and IMF Conference on Embedding Sustainability in Credit Risk Assessment, Yale Initiative on Sustainable Finance Conference, Stanford Institute for Theoretical Economics (SITE) Conference, WE_ARE_IN Macroeconomics and Finance Conference, University of Napoli Climate Finance Workshop, University of Oklahoma and Review of Financial Studies Climate and Energy Finance Research Conference

Seminars: Financial Stability Oversight Council Meetings (Banking Group, Markets Group, Scenario Analysis Group), Federal Reserve Cross-bank Climate Risk Community, Indian Institute of Management Ahmedabad, Central Bank of Brazil, International Monetary Fund, European Central Bank, Federal Reserve Board, Federal Reserve Bank of New York 2023

Conferences: American Economic Association Meeting, System Climate Meeting by the Federal Reserve Bank of San Francisco, Annual Conference of the Central Bank of Chile, Stanford Institute for Theoretical Economics (SITE) Conference, Women Assistant Professors of Finance (WAPFIN) Conference at NYU Stern, Federal Housing Finance Agency (FHFA) Econ Summit, System Banking Conference by the Federal Reserve Board, Financial Stability Conference by the Federal Reserve Bank of Cleveland and the Office of Financial Research

Seminars: Federal Reserve Cross-bank Climate Risk Community, Board of Governor's CREST Seminar, ESSEC-Amundi Chair Webinar, Bank of Korea, Federal Reserve Bank of Richmond 2022

Conferences: NBER International Finance and Macroeconomics Program Meeting, Australasian Finance and Banking Conference, Paris December Finance Meeting, UN PRI Academic Network Conference, OFR Climate Implications for Financial Stability Conference, Cornell University ESG Investing Research Conference, International Journal of Central Banking Research Conference, European Finance Association Meeting, Central Bank Research Association Annual Meeting, Banco de Portugal Conference on Financial Intermediation, OCC Symposium on Climate Risk in Banking & Finance, IMF Macro-Financial Research Conference, System Climate Meeting by the Federal Reserve Bank of San Francisco, NY Fed- Columbia Conference on Environmental Economics and Policy, HEC Paris - CEPR Conference, Banque de France Workshop, Eastern Finance Association Meeting, University of Oklahoma and Review of Financial Studies Climate and Energy Finance Research Conference, European Systemic Risk Board Workshop, Western Economic Association International Meeting (International Banking, Economics, and Finance Association Session)

Seminars: European Central Bank, Central Bank of Chile, Korea University, Hong Kong University of Science and Technology, Vanderbilt Owen, Yale SOM, Federal Reserve Bank of New York, Federal Reserve Board, Stockholm School of Economics, Federal Reserve Bank of Chicago, Warwick Business School, U of SC Darla Moore, Oxford Said, Imperial College London, University of Hong Kong 2021

Conferences: European Conferences of the Economics Community (EC²) Conference, MIT GCFP Conference, Australasian Finance and Banking Conference, IFABS Oxford Conference, International Risk Management Conference, Federal Reserve Stress Testing Conference, Volatility and Risk Institute Conference

Seminars: Society for Financial Econometrics Seminar, NYU Stern, Columbia 2020
GSB (Ph.D.)

Conferences: AFA Ph.D. Poster Session, Federal Reserve Board Pre-Job
Market Conference

ACADEMIC AND PROFESSIONAL SERVICES

Discussions:

ASSA Meeting	2026
(International Banking, Economics, and Finance Association Session)	
NBER Insurance Meeting	2025
Society of Financial Studies Cavalcade	2025
American Finance Association Meeting	2025
American Economic Association Meeting	2024
ECB-NY Fed Workshop	2023
Women in System Economic Research Conference	2023
International Roles of the U.S. Dollar Conference	2023
Paris December Meeting	2022
Southern Economic Association Meeting	2022
Eastern Finance Association Meeting	2022
Australasian Finance and Banking Conference	2021

Refereeing:

Applied Economics Letters, Climate Policy, Ecological Economics, Economic Modeling, Energy Economics, Journal of Banking & Finance, Journal of Corporate Finance, Journal of Economic Surveys, Journal of Finance, Journal of Financial Economics, Journal of International Financial Markets Institutions & Money, Journal of Risk and Insurance, Management Science, Nature Climate Change, Quarterly Journal of Economics, Review of Finance, Review of Financial Studies

Program Committee:

Global Research Forum Conference (by FRB-ECB-NY Fed)	2026
SFS Cavalcade	2026
WE ARE IN Macroeconomics and Finance Conference	2025 – 2026
Financial Management Association Meeting	2025

International Finance and Banking Society Conference	2024, 2026
International Roles of the U.S. Dollar Conference (by FRB-NY Fed)	2024

Session Chair:

Paris December Meeting (Banking/ Financial Intermediation Session)	2022
EC^2 Conference (Macroeconomics of Emissions, Climate Finance Session)	2021
IFABS Oxford Conference (Climate Change Session)	2021

Professional Service:

Research Analyst Coordinator	2025 – Current
Research Analyst and Intern Recruiting Coordinator (Received Vault Award for Excellence)	2023 – 2024, 2026 - 2027
Financial Intermediation Seminar Organizer	2022 – 2023

Pre-doctoral Advising:

Neil Kapoor (Placement: Stanford GSB PhD Program)	2026
Aaron Chen (Placement: Chicago Booth PhD Program)	2025

HONORS AND AWARDS

Best Paper Award, Mid-Atlantic Research Conference	2026
Financial News & KAFA Top-Journal Paper Award	2024
Berkeley Haas Sustainability Research Prize Finalist	2023
Research Grant, National Science Foundation Project “Market Based Climate Stress Tests” (Award # 2218455)	2022
Research Grant, Inter-American Development Bank and Volatility and Risk Institute, NYU Stern	2021
Ph.D. Research Grant, Center for Global Economy and Business, NYU Stern	2020
AFA Ph.D. Student Travel Grant	2020
Jules Bogen Fellowship, NYU Stern	2019 – 2020
NYU Stern Doctoral Fellowship	2015 – 2021
NYU Stern Teaching Commendation	2018
Wharton Undergraduate Research Award	2012

TEACHING EXPERIENCE

Instructor, Foundations of Finance (Undergraduate)	Summer 2018
Overall Evaluation: 5.0/5.0	
<i>Awarded Commendation for Teaching Excellence</i>	
Teaching Fellow, Financial Econometrics (Ph.D.) Prof. Robert Engle	Spring 2018
Teaching Fellow, Volatility (MBA) Prof. Robert Engle	January 2020
Teaching Fellow, Investments (Executive MBA) Prof. Anthony Lynch	Fall 2019
Teaching Fellow, Foundations of Finance (MBA) Prof. Anthony Lynch	Summer 2017 – 2019
Teaching Fellow, Principles of Securities Trading (Undergraduate) Prof. Joel Hasbrouck	Spring & Fall 2017- 2018, 2020
Project Advisor, Stern Signature Project (MBA) Prof. Robert Engle	Spring 2021
Guest Speaker, Climate Finance, NYU Stern, Prof. Johannes Stroebe	Fall 2021
Guest Speaker, Credit Risk and Bankruptcy, NYU Stern, Prof. Viral Acharya	Spring 2022
Guest Speaker, Credit Risk and Bankruptcy, NYU Stern, Prof. Viral Acharya	Spring 2023
Guest Speaker, Credit Risk, NYU Stern, Prof. Viral Acharya	Spring 2023
Guest Speaker, Climate Finance, NYU Stern, Prof. Johannes Stroebe	Fall 2023
Guest Speaker, Climate Finance, Cornell Johnson, Prof. Alissa Kleinnijenhuis	Spring 2024
Guest Speaker, Climate Finance, Stockholm School of Economics, Prof. Ye Zhang	Fall 2024

REFERENCES

Prof. Robert Engle (Co-chair)

New York University, Stern School of Business
Email: rengle@stern.nyu.edu

Prof. Philipp Schnabl (Co-chair)

New York University, Stern School of Business
Email: schnabl@stern.nyu.edu

Prof. Joel Hasbrouck

New York University, Stern School of Business
Email: jhasbrou@stern.nyu.edu

Prof. Ralph Koijen

University of Chicago, Booth School of Business
Email: ralph.koijen@chicagobooth.edu

Prof. Alexi Savov

New York University, Stern School of Business
Email: asavov@stern.nyu.edu

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